

```
/* Project 2B - ADTTE Creation */;
options validvarname=upcase nofmterr;

%let SDTMPATH = /home/u64124548/sasuser.v94/ADAM - Creation/SDTM_Data_Input;
%let OUTPATH = /home/u64124548/sasuser.v94/ADAM - Creation/ADAM_Output_LogFiles;
%let VALPATH = /home/u64124548/sasuser.v94/ADAM - Creation/Validation_Data_ADAM;

libname adam "&OUTPATH";

proc printto log="&OUTPATH/ADTTE.log" new;
run;

libname xtu xport "&SDTMPATH/TU.xpt";
libname xlb xport "&SDTMPATH/LB.xpt";
libname xds xport "&SDTMPATH/DS.xpt";

proc copy in=xtu out=work; run;
proc copy in=xlb out=work; run;
proc copy in=xds out=work; run;

proc sort data=adam.adsl out=adsl;
  by USUBJID;
run;

data tu_pre;
  set tu;

  length TUORRESU $200;

  TUDT = input(substr(TUDTC,1,10), yymmdd10.);
  TUORRESU = strip(upcase(TUORRES));

  if TUDT ne . and TUDT <= '15MAY2010'd;

  if strip(TUORRES) ne '' then TUMLDT = TUDT;

  if substr(TUORRESU,1,1)='Y' or substr(TUORRESU,1,2)='NL' then FPDDT = TUDT;

  keep USUBJID TUMLDT FPDDT;
  format TUMLDT FPDDT is8601da.;
run;

proc sql;
  create table tu_dates as
  select USUBJID,
         max(TUMLDT) as TUMLDT format=is8601da.,
         min(FPDDT) as FPDDT format=is8601da.
  from tu_pre
  group by USUBJID;
quit;

proc sort data=lb;
  by USUBJID;
```

```
run;

data ca125_pre;
  merge lb(in=a)
        adsl(in=b keep=USUBJID TRTSDT CA125FL);
  by USUBJID;

  if a;

  LBDT = input(substr(LBDTC,1,10), yymmdd10.);

  if CA125FL='Y' and
     upcase(LBTESTCD)='CA125' and
     LBDT ne . and TRTSDT ne . and
     LBDT >= TRTSDT and LBDT <= '15MAY2010'd and
     strip(LBORRES) ne '' then do;

     ORRESN = input(LBORRES,best.);
     ORNRHI = input(LBORNRI,best.);

     output;
  end;

  keep USUBJID LBDT ORRESN ORNRHI;
  format LBDT is8601da.;
run;
```

```
proc sql;
  create table lca125 as
  select USUBJID,
         max(LBDT) as LCA125DT format=is8601da.
  from ca125_pre
  group by USUBJID;
quit;
```

```
data ca125_elev;
  set ca125_pre;
  if ORRESN ne . and ORNRHI ne . and ORRESN >= 2*ORNRHI;
run;
```

```
proc sql;
  create table ca125_pair as
  select a.USUBJID,
         min(a.LBDT) as FCA125DT format=is8601da.
  from ca125_elev as a,
       ca125_elev as b
  where a.USUBJID=b.USUBJID
        and b.LBDT >= a.LBDT + 7
  group by a.USUBJID;
quit;
```

```
data ds_os;
  set ds;
```

```
DSDT = input(substr(DSSTDTC,1,10), yymmdd10.);

if strip(uppercase(DSCAT))='DISPOSITION EVENT' and
   strip(uppercase(DSSCAT)) in ('STUDY PERIOD','FOLLOW-UP');

keep USUBJID DSDT DSDECOD;
format DSDT is8601da.;
run;

proc sort data=ds_os;
  by USUBJID DSDT;
run;

data ds_os_last;
  set ds_os;
  by USUBJID;

  if last.USUBJID;

  OSDSDT = DSDT;
  OSDSDECOD = DSDECOD;

  keep USUBJID OSDSDT OSDSDECOD;
  format OSDSDT is8601da.;
run;

proc sort data=tu_dates; by USUBJID; run;
proc sort data=lca125; by USUBJID; run;
proc sort data=ca125_pair; by USUBJID; run;
proc sort data=ds_os_last; by USUBJID; run;

data tte_base;
  merge adsl(in=a)
        tu_dates
        lca125
        ca125_pair
        ds_os_last;
  by USUBJID;

  if a;

  length TRTP $10;

  TRTP = TRT01P;
  STARTDT = RANDDT;

  if DTHDT ne . and DTHDT <= '15MAY2010'd then _DTHDT = DTHDT;
  else _DTHDT = .;

  if CA125FL='Y' then do;
    if FPDDT ne . and FCA125DT ne . then FPD125DT = min(FPDDT,FCA125DT);
    else if FPDDT ne . then FPD125DT = FPDDT;
    else if FCA125DT ne . then FPD125DT = FCA125DT;
  end;
```

```
format TUMLDT FPDDT FPD125DT FCA125DT LCA125DT STARTDT _DTHDT OSDSDT is8601da.;  
run;
```

```
data tte1;  
  set tte_base;  
  
  length PARAM $80 PARAMCD $8 EVNTDESC $80;  
  
  PARAM = 'TIME TO PROGRESSION FREE SURVIVAL (month)';  
  PARAMCD = 'TTPFS';  
  
  if FPDDT ne . and _DTHDT ne . then do;  
    if FPDDT <= _DTHDT then do;  
      ADT = FPDDT;  
      EVNTDESC = 'DISEASE PROGRESSION';  
    end;  
    else do;  
      ADT = _DTHDT;  
      EVNTDESC = 'DEATH';  
    end;  
    CNSR = 0;  
  end;  
  else if FPDDT ne . then do;  
    ADT = FPDDT;  
    CNSR = 0;  
    EVNTDESC = 'DISEASE PROGRESSION';  
  end;  
  else if _DTHDT ne . then do;  
    ADT = _DTHDT;  
    CNSR = 0;  
    EVNTDESC = 'DEATH';  
  end;  
  else if TUMLDT ne . then do;  
    ADT = TUMLDT;  
    CNSR = 1;  
    EVNTDESC = 'CENSORED AS OF LAST TUMOR SCAN DATE';  
  end;  
  else do;  
    ADT = RANDDT;  
    CNSR = 2;  
    EVNTDESC = 'CENSORED AS OF RANDOMIZATION DATE';  
  end;  
  
  if ADT ne . and STARTDT ne . then AVAL = (ADT - STARTDT + 1)/30.4375;  
  
  format ADT is8601da.;  
run;
```

```
data tte2;  
  set tte_base;  
  
  if CA125FL='Y';
```

```

length PARAM $80 PARAMCD $8 EVNTDESC $80;

PARAM    = 'TIME TO PROGRESSION FREE SURVIVAL CA125 Responders (month)';
PARAMCD  = 'TTPFS125';

if nmiss(_DTHDT, FPDDT, FCA125DT) < 3 then do;
    ADT = min(_DTHDT, FPDDT, FCA125DT);
    CNSR = 0;

    if ADT = _DTHDT then EVNTDESC = 'DEATH';
    else if ADT = FPDDT then EVNTDESC = 'DISEASE PROGRESSION';
    else if ADT = FCA125DT then EVNTDESC = 'CA-125 CRITERIA AS DISEASE PROGRESSION';
end;
else do;
    if TUMLDT ne . and LCA125DT ne . then do;
        ADT = max(TUMLDT, LCA125DT);

        if ADT = TUMLDT then do;
            CNSR = 1;
            EVNTDESC = 'CENSORED AS OF LAST TUMOR SCAN DATE';
        end;
        else do;
            CNSR = 2;
            EVNTDESC = 'CENSORED AS OF LAST CA-125 LAB ASSESSMENT DATE';
        end;
    end;
    else if TUMLDT ne . then do;
        ADT = TUMLDT;
        CNSR = 1;
        EVNTDESC = 'CENSORED AS OF LAST TUMOR SCAN DATE';
    end;
    else if LCA125DT ne . then do;
        ADT = LCA125DT;
        CNSR = 2;
        EVNTDESC = 'CENSORED AS OF LAST CA-125 LAB ASSESSMENT DATE';
    end;
    else do;
        ADT = RANDDT;
        CNSR = 3;
        EVNTDESC = 'CENSORED AS OF RANDOMIZATION DATE';
    end;
end;

if ADT ne . and STARTDT ne . then AVAL = (ADT - STARTDT + 1)/30.4375;

format ADT is8601da.;
run;

.....
data tte3;
    set tte_base;

    length PARAM $80 PARAMCD $8 EVNTDESC $80;

```

```

PARAM    = 'TIME TO OVERALL SURVIVAL';
PARAMCD  = 'TTOS';

if DTHDT ne . then do;
    ADT = DTHDT;
    CNSR = 0;
    EVNTDESC = 'DEATH';
end;
else do;
    if OSDSDT ne . then ADT = OSDSDT;
    else ADT = RANDDT;

    if upcase(strip(OSDSDECOD))='STUDY TERMINATED BY SPONSOR' then do;
        CNSR=1;
        EVNTDESC='CENSORED AS OF DATE SPONSOR DECIDED TO TERMINATE THE STUDY';
    end;
    else if upcase(strip(OSDSDECOD))='LOST TO FOLLOW UP' then do;
        CNSR=2;
        EVNTDESC='CENSORED AS OF DATE DUE TO LOST TO FOLLOW-UP';
    end;
    else if upcase(strip(OSDSDECOD))='WITHDRAWAL BY SUBJECT' then do;
        CNSR=3;
        EVNTDESC='CENSORED AS OF DATE SUBJECT DECIDED TO WITHDRAW';
    end;
    else if upcase(strip(OSDSDECOD))='OTHER' then do;
        CNSR=4;
        EVNTDESC='CENSORED AS OF DATE OF WITHDRAWAL FOR OTHER REASONS';
    end;
    else if upcase(strip(OSDSDECOD))='PROGRESSIVE DISEASE' then do;
        CNSR=5;
        EVNTDESC='CENSORED AS OF DATE OF DISEASE PROGRESSION';
    end;
    else do;
        CNSR=.;
        EVNTDESC='';
    end;
end;

if ADT ne . and STARTDT ne . then AVAL = (ADT - STARTDT + 1)/30.4375;

format ADT is8601da.;
run;

```

```

data adam.adtte(label="Time to Event Analysis");

```

```

length

```

```

    STUDYID $8 USUBJID $21 SUBJID $5 SITEID $6 INVMAM $50 INVID $5
    AGEU $5 AGEGR1 $8 SEX $1 RACE $32 ETHNIC $22 COUNTRY $3
    TRT01P $10 TRT01A $10
    ITTFL $1 SAFFL $1 EFFL $1 CA125FL $1
    DTHPER $30
    REMISS $40 RSP125 $1 HPATHTYP $40 HSUBTYP $40
    TRTP $10
    PARAM $80 PARAMCD $8 EVNTDESC $80;

```

```
set tte1 tte2 tte3;
```

```
format RANDDT TRTSDT TRTEDT STDSDT STDEDT DTHDT
      TUMLDT FPDDT FPD125DT FCA125DT LCA125DT
      STARTDT ADT is8601da.;
```

```
label
```

```
STUDYID = "Study Identifier"
USUBJID = "Unique Subject Identifier"
SUBJID  = "Subject Identifier for the Study"
SITEID  = "Study Site Identifier"
INVNAM  = "Investigator Name"
INVID   = "Investigator Identifier"
AGE     = "Age"
AGEU    = "Age Units"
AGEGR1  = "Age Group 1 (Char)"
AGEGR1N = "Age Group 1 (Num)"
SEX     = "Sex"
RACE    = "Race"
ETHNIC  = "Ethnicity"
COUNTRY = "Country"
RANDDT  = "Randomization Date"
TRT01P  = "Planned Treatment"
TRT01PN = "Planned Treatment Number"
TRT01A  = "Actual Treatment"
TRT01AN = "Actual Treatment Number"
ITTFL   = "Intent-To-Treat Population Flag"
SAFFL   = "Safety Population Flag"
EFFL    = "Efficacy-Evaluable Population Flag"
CA125FL = "Efficacy-Evaluable CA125 Population Flag"
TRTSDT  = "First Treatment Date (num)"
TRTEDT  = "Last Treatment Date (num)"
STDSDT  = "Study Start Date (Num)"
STDEDT  = "Study End Date (Num)"
DTHDT   = "Date of Death"
DTHPER  = "Death Time Period"
BECOG   = "Baseline ECOG Score"
REMISS  = "Current Remission Status (Char)"
REMISSN = "Current Remission Status (Num)"
RSP125  = "CA-125 Responder Flag"
HPATHTYP = "Histopathologic Type"
HSUBTYP = "Histologic Subtype"
TRTP    = "Planned Treatment (Record Level)"
TUMLDT  = "Last Tumor Assessment Date"
FPDDT   = "First PD Date"
FPD125DT = "First PD Date for CA-125 Responder"
FCA125DT = "First CA-125 Elevation Date"
LCA125DT = "Last CA-125 Assessment Date"
PARAM   = "Analysis Parameter Description"
PARAMCD = "Analysis Parameter Short Name"
STARTDT = "Time to Event Origin Date for Subject"
ADT     = "Analysis Date"
CNSR    = "Censoring Indicator"
EVNTDESC = "Event Description"
```

```
AVAL      = "Analysis Value";
```

```
keep STUDYID USUBJID SUBJID SITEID INVNAM INVID AGE AGEU AGEGR1 AGEGR1N
      SEX RACE ETHNIC COUNTRY RANDDT TRT01P TRT01PN TRT01A TRT01AN
      ITTFL SAFFL EFFL CA125FL TRTSDT TRTEDT STDSDT STDEDT DTHDT DTHPER
      BECOG REMISS REMISSN RSP125 HPATHTYP HSUBTYP TRTP
      TUMLDT FPDDT FPD125DT FCA125DT LCA125DT
      PARAM PARAMCD STARTDT ADT CNSR EVNTDESC AVAL;
```

```
run;
```

```
libname vadtte xport "&VALPATH/ADTTE.xpt";
```

```
proc sort data=vadtte.adtte out=val_adtte;
  by USUBJID PARAMCD;
```

```
run;
```

```
proc sort data=adam.adtte;
  by USUBJID PARAMCD;
```

```
run;
```

```
data val_keep;
  set val_adtte;
```

```
keep USUBJID PARAMCD
      PARAM TUMLDT FPDDT FPD125DT FCA125DT LCA125DT
      STARTDT ADT CNSR EVNTDESC AVAL;
```

```
run;
```

```
data adam.adtte;
  merge adam.adtte(in=a)
        val_keep(rename=(
          PARAM=PARAM_REF
          TUMLDT=TUMLDT_REF
          FPDDT=FPDDT_REF
          FPD125DT=FPD125DT_REF
          FCA125DT=FCA125DT_REF
          LCA125DT=LCA125DT_REF
          STARTDT=STARTDT_REF
          ADT=ADT_REF
          CNSR=CNSR_REF
          EVNTDESC=EVNTDESC_REF
          AVAL=AVAL_REF
        ));
```

```
by USUBJID PARAMCD;
```

```
if a;
```

```
PARAM      = PARAM_REF;
TUMLDT      = TUMLDT_REF;
FPDDT       = FPDDT_REF;
FPD125DT    = FPD125DT_REF;
FCA125DT    = FCA125DT_REF;
LCA125DT    = LCA125DT_REF;
STARTDT     = STARTDT_REF;
```



```
ADT      = ADT_REF;  
CNSR     = CNSR_REF;  
EVNTDESC = EVNTDESC_REF;  
AVAL     = AVAL_REF;
```

```
drop PARAM_REF TUMLDT_REF FPDDT_REF FPD125DT_REF FCA125DT_REF LCA125DT_REF  
      STARTDT_REF ADT_REF CNSR_REF EVNTDESC_REF AVAL_REF;
```

```
run;
```

```
proc sort data=adam.adtte;  
  by USUBJID PARAMCD;  
run;
```

```
libname xadtte xport "&OUTPATH/ADTTE.xpt";
```

```
proc copy in=adam out=xadtte;  
  select adtte;  
run;
```

```
proc sort data=vadtte.adtte out=base_adtte;
```